

Contact

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Top Skills

Convolutional Neural Networks (CNN)

Large Language Models (LLM)

GitHub

Certifications

Deep Learning using TensorFlow

Data Analysis Using Python

Muhammad Usman Noor

Computer Vision | Machine Learning | ROS2
Lahore, Punjab, Pakistan

Summary

I'm passionate about designing intelligent robotic systems that bridge the physical and digital worlds. With hands-on experience across embedded systems, autonomous navigation, and AI-driven perception, I specialize in building full-stack robotics solutions.

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& : ROS2, SLAM, A*, sensor fusion, Gazebo, RViz

: Jetson Nano/Orin, Pixhawk, LiDAR, depth cameras (D435i)

: YOLOv8, OpenCV, object tracking, stereo depth

& : Supervised learning (scikit-learn, TensorFlow), optimization

& : Exploring LLMs, transformers, prompt engineering, and real-world applications of Generative AI in robotics

& : Python, C++, Flutter, Linux

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Built an for warehouse logistics with person-following, real-time navigation, and safety integration

Developed - using YOLOv8 + RealSense for responsive, autonomous interaction

Currently expanding into to enable smarter, more adaptive robotic behavior

for autonomous navigation and machine learning

on (ros2_robot_slam, Generic-Algorithm,

Titanic_Machine_Learning_Survival)

I'm driven by the mission to combine software intelligence with mechanical systems to solve real-world challenges. Always open to collaboration, learning, and pushing the boundaries of what robots can do.

Let's connect!

Experience

CCRIPT Agency

Computer Vision Engineer

February 2026 - Present (2 months)

Neuralogic

Computer Vision Engineer

February 2026 - Present (2 months)

Upwork

Robotics Developer

September 2025 - Present (7 months)

Pakistan Engineering Council

Registered Engineer

May 2025 - Present (11 months)

Wild Robotics

Software Developer

September 2025 - March 2026 (7 months)

Islamabad

I worked with Wild Robotics on the UR20 robotic arm using ROS2 and Gazebo Harmonic. I developed a custom ROS-based library to accurately simulate vacuum gripper physics in Gazebo Harmonic. I also integrated and configured MoveIt 2 for motion planning, kinematics, and overall robotic arm control.

AA Robotics

Developer and Team lead

June 2022 - October 2024 (2 years 5 months)

Lahore, Punjab, Pakistan

At AA Robotics, I worked on the design and development of drones, unmanned aircraft systems, and ground robots, integrating both hardware and software to create autonomous and semi-autonomous solutions. I utilized ROS (Robot Operating System) for navigation, sensor integration, and mission control, enabling real-time decision-making and precise maneuvering. My contributions included building custom flight controllers, optimizing embedded

systems, and implementing AI-based perception for advanced aerial and ground operations.

Education

National University of Sciences and Technology (NUST)

Master in Robotics & AI, Robotics, and AI · (September 2025 - May 2027)

University of Engineering and Technology, Lahore

B.S.c Mechatronics and Control Engineering , Robotics And AI · (October 2021 - May 2025)